

Size a Saudi data centre

AI Data Center Bootcamp · Week 5 · Day 3 · team case study · Tuesday 15 September 2026

Pick one site. Answer five questions about it with this morning's arithmetic. Show your working, and say what you assumed.

Pick a site

- **HUMAIN's first building:** 50 MW, 18,000 NVIDIA GB300 GPUs. DCD, 28 May 2025; NVIDIA, 13 May 2025.
- **HUMAIN Dammam:** up to 100 MW to start, on land for ten 200 MW buildings. Capacity, 26 Aug 2025; DCD, 28 May 2025.
- **xAI with HUMAIN:** 500 MW for Grok, first cluster about 18,000 GB300 GPUs. CNBC and DCD, 19 Nov 2025.

As of 31 July 2026 no public source confirms either HUMAIN site in service. That is part of the case.

Use these numbers

- Power: PUE 1.25; 10% of IT for switches and storage; 20% headroom.
- Racks: a DGX H100 node is 8 GPUs, 640 GB, 10.2 kW, four to a rack. A GB300 NVL72 rack is 72 GPUs, about 20 TB, about 120 kW.
- Serving: a model's weights at fp8, plus context for 32 conversations of 128K tokens.
- Training: 6 operations per parameter per token, 20 tokens per parameter, 40% of peak. An H100 peaks at 989 TFLOPS.
- Bill: the site draws 65% of its connection on average; USD 0.08 per kWh (30 halalas), or 0.048 at the industrial band.
- Cost: 125 output tokens per second per GPU; USD 450,000 per MW per month for everything but electricity.

The five questions

1. How many racks and GPUs does the site's power buy?
2. What is the largest open model it can serve, and how many copies of it?
3. What is the largest model it could train in six months?
4. What is its electricity bill for a month?
5. What does a million tokens cost, at 30% and at 80% of capacity sold?

Hand in

One page per team on your pod, ~/aidc/site-check.md, by 12:00: the site, the five answers with their working, and one thing the announcement does not tell you. AI for research is fine, but you should handwrite or handtype the final document.